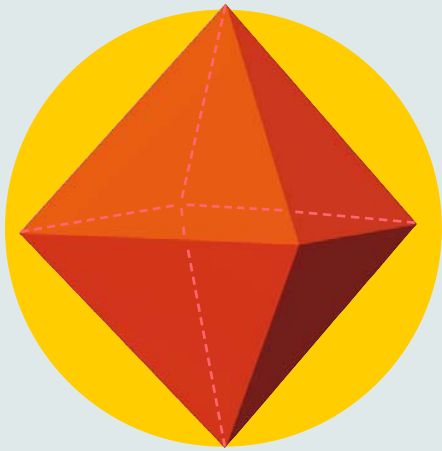


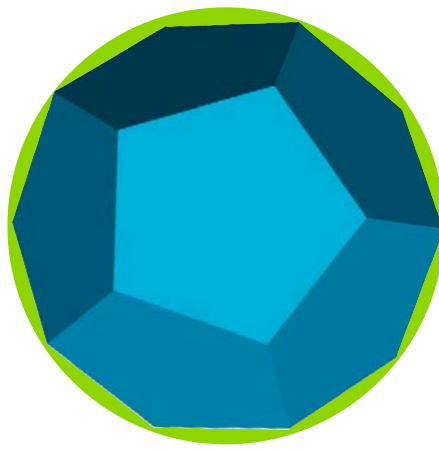
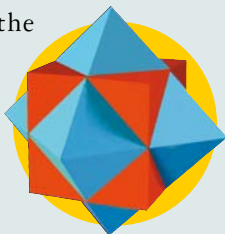


An infinite number of regular polygons can exist in two dimensions, but what if you try making perfectly regular shapes in 3D, each with equal sides, angles, and faces? The ancient Greeks discovered that only five such shapes – called *Platonic solids* – are possible. They thought these perfect shapes were the invisible building blocks of the entire universe.

OCTAHEDRON DODECAHEDRON ICOSAHEDRON



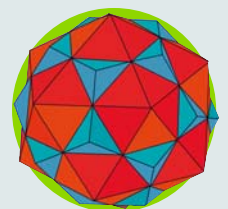
- The octahedron is made of 8 equilateral triangles arranged like two pyramids stuck together.
- Octahedrons, tetrahedrons, and cubes can all completely fill a space, without gaps.
- The Greeks saw the octahedron as being halfway between the tetrahedron (fire) and the cube (earth), so they decided it represented the element of air.
- The octahedron and cube are “dual shapes”. If you chopped all the corners off an octahedron, you’d end up with a cube (and vice versa). If you slotted the two shapes together, the corners of each would stick through the midpoints of the other’s faces.



- A dodecahedron is made of 12 pentagons. In Greek, “dodeca” means 2 + 10.
- The Greeks only needed four shapes for their theory about the elements, and the dodecahedron was the leftover. So as not to leave it out entirely, they decided that the dodecahedron was the shape of the entire universe, with its 12 faces corresponding to the 12 constellations of the zodiac.
- The dodecahedron and icosahedron are dual shapes, just as the cube and octahedron are. If you chopped the corners off a dodecahedron, you’d end up with an icosahedron (and vice versa).



- The icosahedron is made of 20 equilateral triangles.
- This shape represented the element of water to the Greeks. Perhaps they noticed that icosahedrons roll around easily, a bit like flowing water.
- The icosahedron has some surprising connections with nature. Some viruses (including chicken pox) and some microscopic sea creatures have bodies based on an icosahedron.
- If you slotted an icosahedron and dodecahedron together, the corners of each would stick through all the midpoints of the other shape’s faces.



was the shape of the **WHOLE UNIVERSE**